# FamilyVault — Sprint 5 Master Workflow

Milestone Time Capsules, Multi-Trustee Governance & Emergency Red Alert • Serial Deliverable Blueprint • Generated October 01, 2026 • Branch: develop

| | | |
|---|---|---|
| **Sprint Theme:** Milestone Governance & Red Alert | **Lead Architect:** Runeet (realruneett01) | **Target Branch:** develop |
| **Security:** Three-Tier Row Level Security (RLS) | **Verification:** Cryptographic SHA-256 Hashes | **Governance:** Multi-Trustee Quorum Signatures |

## 1. Executive Summary & Sprint Objectives

Sprint 5 operationalizes generational legacy security and life-saving emergency medical response: (1) Milestone-locked time capsules and digital wills remain cryptographically sealed until a scheduled timestamp or multi-trustee quorum is fulfilled. (2) Multi-trustee sign-off API (POST /api/v1/vault/{id}/sign) records verified signatures and unlocks the capsule when quorum is met. (3) The Emergency Red Alert Protocol instantly compiles Eleanor Vance's critical health telemetry (pacemaker specifications, contraindicated allergies) and exposes a 24-hour verifiable offline pass (GET /api/v1/health/emergency-pass/{token}) with an active broadcast trigger.

## 2. Master Chronological Deliverables Sequence by Owner (Serial Dependency Chain)

Each deliverable is executed in strict sequence: completion of deliverable N unlocks deliverable N+1. Below is the exact execution chain across Core Backend, Tree Frontend, Dashboard UI, Graph, and Security:

| Deliverable & Owner | Preceding Work (Must Be Done First) | Scope & Technical Work Executed | Output Artifact / Contract | Next Owner & Work Triggered | Status |
|---|---|---|---|---|---|
| **Deliverable 1**<br>**Runeet**<br>*Core Backend Lead* | PostgreSQL schema with documents table and time_capsule_status enum. | Builds milestone time capsule engine: timestamp unlock triggers, multi-trustee sign-off endpoint POST /api/v1/vault/{id}/sign, and badge query API GET /api/v1/vault/milestones/badges. | endpoints/vault.py<br>schemas/document.py | **Runeet** (Core Backend)<br>Triggered to build Emergency Pass & Alert API. | COMPLETED |
| **Deliverable 2**<br>**Runeet**<br>*Core Backend Lead* | Eleanor Vance emergency medical profile and SHA-256 tamper hash. | Implements scannable 24-hour Emergency Telemetry Pass GET /api/v1/health/emergency-pass/{token} with HMAC/SHA-256 verification and Red Alert Broadcast trigger. | endpoints/health.py<br>test_sprint4_and_sprint5.py | **Anushka** (Tree Frontend)<br>Triggered to render milestone & alert UI. | COMPLETED |
| **Deliverable 3**<br>**Anushka**<br>*Tree Frontend Engineer* | **Deliverables 1 & 2** (Runeet's milestone badge metadata & emergency pass API). | Implements MilestoneNodeBadge.tsx and EmergencyAlertOverlay.tsx on the family tree canvas. Members with locked items show lock pills with signature progress (1/2); distress triggers red beacon. | MilestoneNodeBadge.tsx<br>EmergencyAlertOverlay.tsx | **Raghuvendra** (Dashboard UI)<br>Triggered to wire milestone clocks. | COMPLETED |
| **Deliverable 4**<br>**Raghuvendra**<br>*Dashboard Frontend* | **Deliverable 3** (Anushka's milestone badges and Runeet's vault APIs). | Updates /dashboard/nuclear-family with live milestone countdown clocks, digital will trustee approval status cards, and one-click trustee approval trigger. | nuclear-family/page.tsx | **Riya** (Graph Service Lead)<br>Triggered to map trustee relationships. | COMPLETED |
| **Deliverable 5**<br>**Riya**<br>*Graph Service Lead* | Runeet's trustee relationship data in PostgreSQL. | Creates TRUSTEE_FOR relationship in Neo4j graph, enabling graph traversal queries that identify authorized family trustees and their generational distance from testator. | trustee_graph.cypher | **Ayush** (Security Engineer)<br>Triggered to execute RLS pentest suite. | COMPLETED |
| **Deliverable 6**<br>**Ayush**<br>*Security & QA Lead* | **Deliverables 1 to 5** (All governance and security APIs complete). | Executes multi-tenant breach attack simulation against locked time capsules, unreleased digital wills, and private medical dossiers. Validates zero data leaks across household boundaries. | test_cross_tenant_pentest.py<br>test_rls_policies.py | **Entire Team**<br>Sprint 5 Deliverables Verified & Closed. | COMPLETED |

## 3. Sprint 5 Quality Gate Sign-Off

**Certification:** Multi-trustee quorum governance and emergency telemetry passes validated with zero security leaks. Time capsule release logic preserves strict private isolation until quorum completion.
**Signed:** Runeet (Lead Architect & Core Backend) • Ayush (Lead QA)